

Golden Examples

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Domain: Combinatorics

Date: April 13, 2026

Paper: <https://arxiv.org/pdf/2603.06269>

Prompt: Let $\mathfrak{S}_n$ be the set of all permutations of $[n]$, and let $\pi \in \mathfrak{S}_n$. We say that the one-line notation of $\pi = \pi_1\pi_2\ldots\pi_n$ (where $\pi_i = \pi(i)$ for each $i \in \{1, 2, \ldots, n\}$) is σ -avoiding if π does not contain a subsequence $\pi_{i_1}, \pi_{i_2}, \ldots, \pi_{i_k}$ (with $1 \leq i_1 < i_2 < \cdots < i_k \leq n$) that is order-isomorphic to a given permutation $\sigma = \sigma_1\sigma_2\ldots\sigma_k$. We say that π contains σ if there is a subsequence $\pi_{i_1}, \pi_{i_2}, \ldots, \pi_{i_k}$ (with $1 \leq i_1 < i_2 < \cdots < i_k \leq n$) that is order-isomorphic to a given permutation $\sigma = \sigma_1\sigma_2\ldots\sigma_k$.

A cyclic permutation of $[n]$ is a permutation of $[n]$ that is composed of a single n -cycle when written as a product of disjoint cycles. Every cyclic permutation π can be written either in one-line notation as $\pi = \pi_1\pi_2\ldots\pi_n$ where $\pi_i = \pi(i)$ for each $i \in \{1, 2, \ldots, n\}$ or in cycle form $\pi = (c_1, c_2, \ldots, c_n)$, where $c_k = \pi(c_{k-1})$ for $k > 1$ and $c_1 = \pi(c_n)$. If $c_1 = 1$, then it is called the standard cycle form of π , which we denote by $C(\pi)$. That is, $C(\pi) = (1, c_2, \ldots, c_n)$.

We say that the cyclic rotation $(c_i, c_{i+1}, \ldots, c_n, c_1, c_2, \ldots, c_{i-1})$ avoids the pattern σ if $c_i c_{i+1} \ldots c_n c_1 \ldots c_{i-1}$ avoids σ .

We say that all cycle forms of π avoid a pattern τ if every cyclic rotation of $C(\pi)$ avoids τ .

Let $n = 26$. Let N be the number of cyclic permutations $\pi \in \mathfrak{S}_n$ satisfying all of the following conditions: 1. All cycle forms of π avoid the pattern 1432. 2. In the standard cycle form $C(\pi) = (1, c_2, \ldots, c_n)$, $c_n \neq n$ and $c_2 = 6$. 3. The one-line notation of π avoids the pattern 4321. 4. The one-line notation of π contains the pattern 321.

Find $N \pmod{1000}$.

Explanation: Lemma 3.4 gives us the count of cyclic permutations of $[n]$ such that the one line notation avoids $\delta_4 = 4321$ and all cyclic rotations of $C(\pi)$ avoid 1432.

Lemma 3.4 (i) and (ii) cover the cases when $c_n = n$, which we do not want.

Lemma 3.4 (iii) and (iv) cover the cases when $c_n \neq n$.

Lemma 3.4 (iii) covers the case when $C(\pi) = (1, j, c_3, \ldots, n, 2, 3, \ldots, j-1, \ldots, c_n)$. It tells us that all letters between j and n are in increasing order, as well as the letters after $j-1$. Also, it tells us that, for $n = 26$, there are $2^{26-j-1} = 2^{25-j}$ such cyclic permutations.

Lemma 3.4 (iv) covers the case when $C(\pi) = (1, j, c_3, \ldots, 2, \ldots, i, \ldots, n, i+1, i+2, \ldots, j-1)$ for some $i \in [2, j-2]$. The lemma tells us that the letters in $[j+1, n-1]$ occur between j and n , and are in increasing order. Also, it tells us that, for $n = 26$, there are $\binom{26-3}{j-3} - 1 = \binom{23}{j-3} - 1$ such cyclic permutations.

Condition 2 tells us that $j = 6$.

So the total number of cyclic permutations satisfying conditions 1, 2 and 3 is $2^{25-6} + \binom{23}{6-3} - 1 = 526058$.

Now, we need to subtract the number of cyclic permutations in the above count that avoid the pattern 321 (Condition 4), which corresponds to $a_{26,6}^\circ(\delta_3; 1432)$.

The proof of Lemma 2.7 tells us that for each j ($4 \leq j \leq n$), there is exactly one valid permutation that avoids δ_3 if $2j - 3 \leq n$. Since in our case $j = 6$ and $n = 26$, this inequality is satisfied. Hence, $a_{26,6}^\circ(\delta_3; 1432) = 1$. Note that the proof of Lemma 2.7 says that this unique permutation ends in $j - 1 = 5 \neq 26$, so it is included in our above count.

Thus, the total count is $N = 526058 - 1 = 526057$.

Thus, $N \pmod{1000} = 57$.

Domain: Graph Theory

Date: June 21, 2026

Paper: <https://arxiv.org/pdf/2401.12347>

Prompt: Let $N = 2^5 3^3$. For each positive divisor d of N , let $v_2(d)$ denote the exponent of 2 in d , and define $k(d) = v_2(d) + 1$. Now define the distance set $D_d = d \cdot \{1, 2, \dots, 2k(d) - 1, 2k(d) + 1, 4k(d)\}$, and let $G_d = G(\mathbb{Z}, D_d)$ be an integer distance graph. Let $\chi(G)$ denote the chromatic number of the graph G , and $\omega(G)$ the clique number of G . What is the exact value of $P = \frac{1}{7} \prod_{d|N} \frac{\chi(G_d)}{\omega(G_d)}$? Final answer format: return a fully reduced rational fraction without LaTeX code. Example: 7/12

Explanation: Let

$$S_k = \{1, 2, \dots, 2k - 1, 2k + 1, 4k\}.$$

Then the distance set in the problem is

$$D_d = dS_{k(d)}, \quad k(d) = v_2(d) + 1.$$

The paper notes in the introduction (4th paragraph of page 2) that if every element of a distance set is divisible by d , then

$$G(d_1, d_2, \dots, d_m)$$

is isomorphic to d disjoint copies of

$$G\left(\frac{d_1}{d}, \frac{d_2}{d}, \dots, \frac{d_m}{d}\right).$$

Therefore,

$$\chi(G(\mathbb{Z}, D_d)) = \chi(G(\mathbb{Z}, S_{k(d)})),$$

and

$$\omega(G(\mathbb{Z}, D_d)) = \omega(G(\mathbb{Z}, S_{k(d)})).$$

So the divisor d matters only through

$$k(d) = v_2(d) + 1.$$

Every positive divisor of

$$N = 2^5 3^3$$

has the form

$$d = 2^a 3^b, \quad 0 \leq a \leq 5, \quad 0 \leq b \leq 3.$$

Thus

$$k(d) = v_2(d) + 1 = a + 1.$$

Apply Theorem 2.2 of the paper. We have

$$|D| = (2k - 1) + 2 = 2k + 1.$$

Therefore, for the non-exceptional cases covered by the theorem, namely the cases $k \geq 2$ needed here, we get

$$\chi(G(\mathbb{Z}, S_k)) = 2k + 2,$$

and

$$\omega(G(\mathbb{Z}, S_k)) = 2k.$$

So for $k = 2, 3, 4, 5, 6$,

$$\frac{\chi(G(\mathbb{Z}, S_k))}{\omega(G(\mathbb{Z}, S_k))} = \frac{2k + 2}{2k} = \frac{k + 1}{k}.$$

We need to consider the case $k = 1$ separately. When $k = 1$,

$$S_1 = \{1, 3, 4\}.$$

If one blindly inserted $k = 1$ into the formula from Theorem 2.2, one would get

$$\omega(G(\mathbb{Z}, S_1)) = 2k = 2.$$

But that is false. Indeed, the vertices

$$\{0, 1, 4\}$$

form a triangle, because their pairwise distances are

$$1, \quad 3, \quad 4,$$

all of which belong to S_1 . The referenced paper, in its proof of Theorem 2.2, specifically mentions that the distance set is none of the exceptional Kemnitz and Marangio clique-producing type, which excludes the case $S_1 = \{1, 3, 4\}$. In fact, in the introduction of the referenced paper (page 2), it recalls the known 3-element classification, including the case of the distance set of the form $\{x, y, x + y\}$ with $x \not\equiv y \pmod{3}$, which applies to S_1 with $x = 1, y = 3$.

One can easily find that

$$\omega(G(\mathbb{Z}, S_1)) = 3,$$

and

$$\chi(G(\mathbb{Z}, S_1)) = 4.$$

Therefore

$$\frac{\chi(G(\mathbb{Z}, S_1))}{\omega(G(\mathbb{Z}, S_1))} = \frac{4}{3}.$$

For each fixed a , the corresponding value of k is

$$k = a + 1.$$

Each such a occurs 4 times because b can be 0, 1, 2, 3. Hence

$$P = \frac{1}{7} \left(\frac{4}{3} \cdot \frac{3}{2} \cdot \frac{4}{3} \cdot \frac{5}{4} \cdot \frac{6}{5} \cdot \frac{7}{6} \right)^4 \cdot \frac{1}{7} \cdot \frac{38416}{81} = \frac{5488}{81}.$$

Domain: Analytic Number Theory

Date: April 15, 2026

Paper: <https://arxiv.org/pdf/1102.3236>

Prompt: Let f be the multiplicative function supported on the squarefree numbers and defined as follows on prime numbers p :

$$f(p) = 1 \text{ when } p \equiv 1, 3, 7 \pmod{17}, \quad f(p) = 0 \text{ else.}$$

Further, let $\psi(x) = [x / (\sum_{n \leq x} f(n))]^{1/13}$ for $x \geq 100$.

Next, we define $r_j = \exp(\psi(x)^j)$ for each index $0 \leq j \leq 5$ and introduce the real number α implicitly defined by the identity

$$\sum_{j=1}^5 (7-j)^\alpha \log(7-j) = \sum_{j=1}^5 (6-j).$$

Let $A(x)$ count the natural numbers $n \leq x$ which have at least one divisor of the form $d_1 d_2 d_3 d_4 d_5 | n$ satisfying $d_j \in (r_j, 2r_j]$ for each index $j = 1, \dots, 5$. Calculate the limit

$$\lim_{x \rightarrow \infty} \frac{\log \left(\left[\frac{A(x) \psi(x)^{5 - \sum_{j=1}^5 (7-j)^\alpha}}{x} \right] \right)}{\alpha \log \log x}.$$

The answer is a ratio of two relatively prime integers with positive denominator. If the answer is negative, put the negative sign in front with no space.

Explanation: Let us first prove an asymptotic for $\psi(x)$. Here Wirsing's Theorem is applicable since f is a positive multiplicative function for which $f(p) \leq 1$ at all primes p and

$$\sum_{p \leq x} \frac{f(p) \log p}{p} \sim \frac{3}{\varphi(17)} \log x$$

as $x \rightarrow \infty$, which can be seen by adding the contributions from each of the three residue classes $1, 3, 7 \pmod{17}$. We gather that

$$\sum_{n \leq x} f(n) \sim c \frac{x}{\log x} \prod_{p \leq x} \left(1 + \frac{f(p)}{p} \right) \sim c' \frac{x}{(\log x)^{13/16}}$$

for some absolute constants $c, c' > 0$, since the logarithm of the Euler product just above is asymptotically equivalent to

$$\sum_{p \equiv 1, 3, 7 \pmod{17}} \frac{1}{p} \sim \frac{13}{16} \log \log x + \kappa + o(1)$$

by Mertens' Theorem (in arithmetic progressions). It follows that $\psi(x) \sim c''(\log x)^{1/16}$.

In order to approximate $A(x)$, we require Theorem 1.3 of Koukoulopoulos's article (with $k = 5$). Writing $F(t) = t \log t - t + 1$ for $t > 0$, the theorem asserts that

$$\begin{aligned} A(x) &\asymp \frac{x}{\sqrt{\log \log r_5}} \prod_{j=1}^5 \left(\frac{\log r_j}{\log r_{j-1}} \right)^{-F((7-j)^\alpha)} \\ &\asymp \frac{x}{\sqrt{\log \log x}} \exp \left(-l(x) \sum_{j=1}^5 \left[\alpha(7-j)^\alpha \log(7-j) - (7-j)^\alpha + 1 \right] \right) \\ &= \frac{x}{\sqrt{\log \log x}} \exp \left(-l(x) \sum_{j=1}^5 \left[\alpha(6-j) - (7-j)^\alpha + 1 \right] \right), \end{aligned}$$

where we have written $l(x) = \log(\psi(x))$ and used that $\log r_j / \log r_{j-1} = \psi(x)$. We gather that

$$\frac{\log \left(\left[\frac{A(x)\psi(x)^{5-\sum_{j=1}^5 (7-j)^\alpha}}{x} \right] \right)}{\alpha \log \log x} \sim -\frac{l(x) \sum_{j=1}^5 (6-j)}{\log \log x} \sim -15/16.$$

Domain: Applied Math

Date: May 24, 2026

Paper: <https://arxiv.org/pdf/2506.21120>

Prompt: Consider the following discrete equation for real q_n on the grid $x = n\epsilon$ with $\epsilon > 0$ and $n \in \mathbb{Z}$

$$\frac{\mu}{\epsilon^2} (q_{n+2} - 2q_n + q_{n-2}) + \frac{1}{\epsilon^2} (q_{n+1} - 2q_n + q_{n-1}) + (|q_n|^2 - 1) q_n = 0.$$

Set $z := n\epsilon$ and $q(z) := q_n$ and expand the solution in the perturbation series

$$q(z) = \sum_{j=0}^{\infty} q_j(z) \epsilon^{2j}, \quad \epsilon \rightarrow 0^+.$$

Assume that $q_0(0) = \sqrt{2}$ and $q(z)$ vanishes at $z = \pm\infty$ and all perturbation coefficients are even in z . By $z_s \in \mathbb{C}$ denote the principal singularity of $q_0(z)$. Let A be the coefficient of $(z - z_s)^{-2}$ in the Laurent expansion of $q_1(z)$ for $z \rightarrow z_s$.

Assume the factorial-over-power ansatz, that is

$$q_n(z) \sim \frac{U(z) \Gamma(2n + \gamma)}{\chi(z)^{2n + \gamma}}, \quad n \rightarrow \infty,$$

with the amplitude U , the singulant χ , and the shift $\gamma \in \mathbb{R}$. The singulant χ is associated with the singularity z_s and normalized by $\chi(z_s) = 0$ (its branch is the canonical one obtained by analytic continuation from z_s to the real line along the straight path from z_s to 0). The shift γ is uniquely defined as the real number for which the following limit exists, is finite, and does not vanish identically in $\mathbb{R}$

$$U(z) = \lim_{j \rightarrow \infty} \frac{q_j(z) \chi(z)^{2j + \gamma}}{\Gamma(2j + \gamma)}.$$

Also, by ζ denote the Riemann zeta function and

$$I(\alpha) := \int_0^\infty \frac{(\ln x)^2 \arctan x}{x^2 - 2x \cos \alpha + 1} dx, \quad 0 < \alpha < 2\pi.$$

Answer Format: Return only a fully reduced rational fraction as a LaTeX expression with no spaces. Example: $\frac{7}{12}$

What is the value of

$$\frac{\gamma}{4} + \frac{\pi^{-4} \gamma^2 I\left(\frac{\pi}{6}\right)}{11A},$$

when

$$\mu = \frac{240 - 17\sqrt{2}\pi}{16(17\sqrt{2}\pi - 60)}?$$

Explanation: In the limit $\epsilon \rightarrow 0^+$ the PDE can be expanded into the formal power series in ϵ

$$2 \sum_{i=0}^{\infty} \frac{1}{(2i)!} (\mu 2^{2i} + 1) \epsilon^{2i-2} y^{(2i)}(z) + (|q(z)|^2 - 1) q(z) = 0,$$

where $q^{(i)}$ denotes the i -th derivative. Since we are interested only in q_0 and q_1 we can rewrite the equation and extract the relevant orders in ϵ . That is to say, we have

$$\beta^2 q'' + \epsilon^2 \frac{4\mu + 1}{12} q^{(4)} + q^3 - q = O(\epsilon^3),$$

where $\beta = \sqrt{4\mu + 1}$. Now, plugging the perturbation series into the above we obtain the leading order equation as

$$\beta^2 q_0'' + q_0^3 - q_0 = 0,$$

which has the famous soliton solution

$$q_0(z) = \sqrt{2} \operatorname{sech} \left(\frac{z}{\beta} \right),$$

where we used our condition that $q_0(0) = \sqrt{2}$. This function has singularities at

$$\beta \left(\frac{\pi}{2} i + k\pi i \right),$$

from which the principal is the one with $k = 0$. In exactly the same way, we can obtain the differential equation for the next order, namely

$$\beta^2 q_1'' + 3q_0^2 q_1 - q_1 = -\frac{16\mu + 1}{12} q_0^{(4)}.$$

This equation is of the form $Lq_1 = -\beta q_0^{(4)}$. Therefore, we can look for a solution expressed in terms of q_0 , q_0^3 , and zq_0' . After some algebra, we obtain

$$q_1(z) = \frac{16\mu + 1}{24(4\mu + 1)^2} (4q_0(z)^3 - 7q_0(z) - zq_0'(z)).$$

The uniqueness of this solution follows from the boundedness and localization at $z = 0$ (also from the given condition). We are looking for the coefficient of $(z - z_0)^{-2}$. Two terms: q_0^3 and zq_0' contribute with the terms $(z - z_s)^{-1}$ and $(z - z_s)^{-3}$, respectively, so looking for $(z - z_s)^{-2}$ we have to focus only on the term zq_0'

$$q_0(z) = -\frac{i\sqrt{2(1+4\mu)}}{z - z_s} + O(z - z_s), \quad z \rightarrow z_s.$$

Plugging that into the formula for q_1 and expanding into Laurent series, we obtain

$$A = \frac{\pi(16\mu + 1)}{24\sqrt{2}(4\mu + 1)} = \frac{5}{17},$$

where we have used the given value for μ .

Now, we proceed to the late-time asymptotic analysis. First, matching the $O(\epsilon^{2j})$ terms, we obtain the general equation for the iterate

$$2 \sum_{i=1}^{j+1} \frac{1 + 2^{2i}\mu}{(2i)!} q_{j-i+1}^{(2i)} + (3q_0^2 - 1)q_j = F(q_0, \dots, q_{j-1}),$$

for some F . For large n , the factorial-over-power ansatz is

$$q_n(z) \sim \frac{U(z)\Gamma(2n + \gamma)}{\chi(z)^{2n+\gamma}},$$

where the amplitude U , the singulant χ , and the shift γ have to be determined by plugging the ansatz into the formal infinite-order equation and matching the first terms. Note that, when we differentiate the ansatz each derivative brings a factor of $-(2j)\chi'/\chi$, hence

$$q_{j-i+1}^{(2i)} \sim U(z)(\chi')^{2i} \frac{\Gamma(2j + \gamma + 2)}{\chi(z)^{2j+\gamma+2}}.$$

Thanks to that, when plugged back into the differential equation, each term scales in the same way as $j \rightarrow \infty$. Hence, its balance with the nonlinear term $(3q_0^2 - 1)q_j$ gives

$$\frac{\Gamma(2j + \gamma + 2)}{\chi^{2j+\gamma+2}} \sim \frac{\Gamma(2j + \gamma)}{\chi^{2j+\gamma}}.$$

But $\Gamma(2j + \gamma)/\Gamma(2j + \gamma + 2) \sim (2j)^{-2}$ so that the nonlinear term is subdominant and can be dropped in the balance. We then arrive at

$$2U(z) \frac{\Gamma(2j + \gamma + 2)}{\chi^{2j+\gamma+2}} \sum_{i=1}^{\infty} \frac{1 + 2^{2i}\mu}{(2i)!} (\chi')^{2i} = 0.$$

And by splitting these two sums we recognize the hyperbolic cosine series! That is,

$$\cosh(\chi') - 1 + \mu(\cosh(2\chi') - 1) = 0.$$

Using the formula for the cosh of a double argument, we solve a quadratic yielding

$$\cosh \chi' = \frac{-1 \pm (1 + 4\mu)}{4\mu}.$$

Therefore, the derivative of the singulant is constant so that we obtain two families of solutions

$$\chi_{1,\pm}(z) = \pm 2\pi i(z - z_s), \quad \chi_{2,\pm}(z) = \ln \left(-\frac{1}{2\mu}(1 + 2\mu \pm \beta) \right) (z - z_s),$$

where we used the condition for the singulant to vanish at the singular point of the leading-order solution (this follows from the standard theory of exponential asymptotics).

Now, we turn to the amplitude. From the main equation, we have to extract the derivative term U' . The algebra gives the result that a balance in the $O(q_{j+1/2})$ term gives an equation

$$2U'(z) \sum_{i=1}^{j-1} \frac{(1 + 2^{2i}\mu)(\chi')^{2i-1}}{(2i-1)!} = 0,$$

which neatly sums to

$$2U'(z) (2\mu \sinh(2\chi') + \sinh(\chi')) = 0.$$

Note that plugging $\chi'_{2,\pm}$ into the above forces $U' = 0$, but plugging $\chi'_{1,\pm}$ leaves U' undetermined. Therefore, we have to move to the higher order balance. This is,

$$U''(z) \sum_{i=1}^{j-1} \frac{(1 + 2^{2i}\mu)(\chi')^{2i-2}}{(2i-2)!} + (3q_0^2 - 1)U = 0.$$

Plugging $\chi'_{1,\pm}$ and summing the series finally yields the ODE for the amplitude

$$\beta^2 U'' + \left(6 \left(\operatorname{sech} \left(\frac{z}{\beta} \right) \right)^2 - 1 \right) U = 0.$$

This equation is known in the literature and has hyperbolic solutions

$$U(z) = K_1 U_1(z) + K_2 U_2(z),$$

where

$$U_1(z) = \frac{1}{2} \tanh \left(\frac{z}{\beta} \right) \operatorname{sech} \left(\frac{z}{\beta} \right), \quad U_2(z) = \operatorname{sech} \left(\frac{z}{\beta} \right) \left(\frac{6z}{\beta} \tanh \left(\frac{z}{\beta} \right) + \cosh \left(\frac{2z}{\beta} \right) - 5 \right).$$

By expanding our amplitude near the leading-order singularity, we obtain

$$U(z) \sim \Lambda_1 (z - z_s)^3 + \frac{\Lambda_2}{(z - z_s)^2}, \quad z \rightarrow z_s,$$

for some constants $\Lambda_{1,2}$. Now, by plugging this form of the amplitude into the factorial-over-power ansatz we notice that as $j \rightarrow \infty$ the term $(z - z_s)^{-2}$ is subdominant and with $\chi(z) \sim 2\pi i(z - z_s)$ and $U(z) \sim \Lambda_1 (z - z_s)^3$, we obtain

$$q_j(z) \sim \frac{\Lambda_1 (z - z_s)^3 \Gamma(2j + \gamma)}{(2\pi i(z - z_s))^{2j + \gamma}}.$$

But these asymptotics have to be consistent with the leading-order! For $j = 0$ the above is

$$q_0(z) \sim (z - z_s)^{3-\gamma},$$

while the leading order term

$$q_0(z) = \sqrt{2} \operatorname{sech} \left(\frac{z}{\beta} \right) \sim (z - z_s)^{-1}, \quad z \rightarrow z_s.$$

Matching these two yields

$$3 - \gamma = -1 \quad \rightarrow \quad \gamma = 4.$$

The next part is the computation of the given integral $I(\alpha)$. The denominator is standard (as in Poisson kernel) and has a certain symmetry under the transformation $x \mapsto x^{-1}$, for which

$$I(\alpha) = \int_0^\infty \frac{(\ln x)^2 \left(\frac{\pi}{2} - \arctan x \right)}{1 - 2x \cos \alpha + x^2} dx.$$

Note that the part with $\arctan x$ is actually $-I(\alpha)$. Hence,

$$I(\alpha) = \frac{\pi}{4} J(\alpha), \quad J(\alpha) := \int_0^\infty \frac{(\ln x)^2}{x^2 - 2x \cos \alpha + 1} dx.$$

The computation now boils down to finding $J(\alpha)$. To this end, we use the Feynman trick, that is, differentiation with respect to the parameter. Introduce the function

$$G(t) := \int_0^\infty \frac{x^t}{x^2 - 2x \cos \alpha + 1} dx.$$

Then, $J(\alpha) = G''(0)$. But the integral for G is known and can be computed via the residue technique,

$$G(t) = \frac{\pi \sin((\pi - \alpha)t)}{\sin(\pi t) \sin \alpha}.$$

By Taylor expansion, we easily compute the second derivative and hence,

$$G''(0) = \frac{\alpha(\pi - \alpha)(2\pi - \alpha)}{3 \sin \alpha}.$$

From which it follows that

$$I(\alpha) = \frac{\pi \alpha(\pi - \alpha)(2\pi - \alpha)}{12 \sin \alpha}.$$

Finally, we have

$$I\left(\frac{\pi}{6}\right) = \frac{55\pi^4}{1296},$$

and the final answer is

$$\frac{\gamma}{4} + \frac{\pi^{-4} \gamma^2 I\left(\frac{\pi}{6}\right)}{11A} = \frac{98}{81}.$$

Domain: Algebraic Geometry

Date: May 13, 2026

Paper: <https://arxiv.org/pdf/math/0308036>

Prompt: We say a Noetherian integral $\mathbb{Q}$ -scheme S is admissible if the topological space underlying S has exactly 3 points and the function field of S has at most countable transcendence degree over $\mathbb{Q}$. If S is admissible with category of coherent sheaves $\mathfrak{Coh}(S)$, then for each $\mathcal{E} \in \mathfrak{Coh}(S)$ we get a contravariant functor $F_{\mathcal{E}} : \mathbf{Sch}/S \rightarrow \mathbf{Set}$ given by

$$F_{\mathcal{E}} : (\varphi : T \rightarrow S) \mapsto H^0(T, \varphi^* \mathcal{E}).$$

For each admissible S , we define an associated topological space $\mathfrak{X}(S)$ as follows: the set underlying $\mathfrak{X}(S)$ is

$$\mathfrak{X}(S) := \{F_{\mathcal{E}}(-) : \mathcal{E} \in \mathfrak{Coh}(S), F_{\mathcal{E}} \text{ is representable by an } S\text{-scheme}\} / \sim,$$

where the relation $\sim$ is natural isomorphism of functors. To topologize $\mathfrak{X}(S)$, by construction every $F_{\mathcal{E}}(-) \in \mathfrak{X}(S)$ admits a fine moduli space $X_{F_{\mathcal{E}}}$, unique up to isomorphism of S -schemes. We give $\mathfrak{X}(S)$ the initial topology with respect to the map $\mathfrak{X}(S) \rightarrow \mathbb{Z}$ given by

$$F_{\mathcal{E}} \mapsto \dim X_{F_{\mathcal{E}}}.$$

This is an absolute dimension (not relative to S), and the codomain is discrete.

Let $\mathcal{S}$ be the set of homeomorphism classes of topological spaces X of the form

$$X \cong \prod_{i \in I} \mathfrak{X}(S_i)^{\eta_i},$$

for distinct admissible schemes S_i and positive integers η_i , with finite nonempty index set I . The set $\mathcal{S}$ is naturally a semigroup under product of topological spaces, and we let R denote the semigroup ring corresponding to $\mathcal{S}$ over $\mathbb{Q}$. How many connected components does $\mathrm{Spec}(R)$ have?

Explanation: The answer is 3. We will prove a number of results leading to this, some of which are perhaps stronger than necessary. We will start with the following:

Lemma 1: For an admissible scheme S , the following are equivalent:

1. S is affine
2. S is separated
3. $\mathrm{Pic}(S) = 0$.

Proof: The biconditional (2) $\iff$ (1) follows from the fact that any Noetherian scheme with finitely many points is separated if and only if it is affine. For (1) $\implies$ (3), if S is affine, then $R := \Gamma(S, \mathcal{O}_S)$ is a semilocal Noetherian integral domain, and Serre's splitting theorem, combined with Forster's theorem, imply that any finitely generated projective module over R is free. It follows that $\text{Pic}(S) = 0$.

The last implication, (3) $\implies$ (1), is a bit more involved. We will actually prove (3) $\implies$ (2), which is sufficient.

Let S be an admissible scheme that is not separated. The topological space underlying S must be the categorical pushout $\mathbb{S} \leftarrow \{*\} \rightarrow \mathbb{S}$, where $\mathbb{S}$ is the Sierpinski space and the maps $\{*\} \rightarrow \mathbb{S}$ are the embeddings of the unique open point. To see this, note that there are, up to homeomorphism, only 5 topological spaces of cardinality 3 that are T_0 . Amongst these, two are not connected and hence not capable of supporting an integral scheme structure, and another two contain a point whose only open neighborhood is the whole space (and hence any scheme structure on them is affine). The only remaining option is the space previously described.

Any scheme with at most two points is affine. It follows that S admits an affine open cover $U, V \subseteq S$ with $|U| = |V| = 2$. Writing $U \cong \text{Spec}(A)$ and $V \cong \text{Spec}(B)$ for Noetherian integral $\mathbb{Q}$ -algebras A, B , we apply Mayer-Vietoris in sheaf cohomology to the sheaf of units $\mathcal{O}_S^\times$ to obtain an exact sequence

$$0 \rightarrow H^0(S, \mathcal{O}_S^\times) \rightarrow H^0(U, \mathcal{O}_S^\times) \oplus H^0(V, \mathcal{O}_S^\times) \rightarrow H^0(U \cap V, \mathcal{O}_S^\times) \rightarrow H^1(S, \mathcal{O}_S^\times) \rightarrow 0.$$

Here, the higher cohomology vanishes because the rings A, B are local, so Kaplansky's theorem for projective modules over local rings implies

$$H^1(U, \mathcal{O}_S^\times) \cong \text{Pic}(A) = 0,$$

and similarly for $V \cong \text{Spec}(B)$. Writing $\text{Frac}(A) \cong K \cong \text{Frac}(B)$, the exact sequence above is

$$0 \rightarrow H^0(S, \mathcal{O}_S^\times) \rightarrow A^\times \times B^\times \rightarrow K^\times \rightarrow \text{Pic}(S) \rightarrow 0.$$

It suffices, then, to show that the map $A^\times \times B^\times \rightarrow K^\times$ is not surjective. This is immediate, however: if this map were surjective, then the following map of rings would also be surjective:

$$A \otimes_{\mathbb{Z}} B \cong H^0(U, \mathcal{O}_S) \otimes_{\mathbb{Z}} H^0(V, \mathcal{O}_S) \rightarrow H^0(U \cap V, \mathcal{O}_S) \cong K.$$

If this were surjective, then it would follow immediately that S is separated, by Stacks 26.21.7. This is a contradiction, so we're done. $\square$

We now characterize the cardinality of the set of isomorphism classes of locally free sheaves of rank n over a non-affine admissible S :

Lemma 2: Let n be a positive integer and let S be a non-affine admissible scheme. Then there are exactly $\aleph_0$ isomorphism classes of locally free coherent sheaves over S of rank n .

Proof: For $n = 1$, we need to prove that $|\mathrm{Pic}(S)| = \aleph_0$. We evidently have $|\mathrm{Pic}(S)| \leq \aleph_0$ since, if K is the function field of S , there is a surjection $K^\times \rightarrow \mathrm{Pic}(S)$, and $K/\mathbb{Q}$ has at most countable transcendence degree.

For the other direction, take the same affine open cover $U, V \subseteq S$ from *Lemma 1* and let $R \subseteq K$ denote the image of the natural map

$$H^0(U, \mathcal{O}_S) \otimes_{\mathbb{Z}} H^0(V, \mathcal{O}_S) \rightarrow K.$$

Writing $U \cong \mathrm{Spec}(A), V \cong \mathrm{Spec}(B)$ as before, there is then a containment

$$\mathrm{im}(A^\times \times B^\times \rightarrow K^\times) \subseteq R^\times,$$

so that Mayer-Vietoris gives a surjection

$$\mathrm{Pic}(S) \rightarrow K^\times / R^\times.$$

It then suffices to prove that the codomain is infinite, but this is immediate: if it were finite, then the ring extension $R \rightarrow \mathrm{Frac}(R) \cong K$ would be integral, implying R is already a field and hence that the natural map $A \otimes_{\mathbb{Z}} B \rightarrow K$ is surjective. Again by Stacks 26.21.7, this would mean S is separated, but by *Lemma 1*, it isn't. This concludes the proof for the $n = 1$ case.

For $n > 1$, let $\mathcal{A}(n)$ denote the set of isomorphism classes of rank n locally free sheaves over S . Consider the map $\mathrm{Pic}(S) \rightarrow \mathcal{A}(n)$ given by

$$\mathcal{L} \mapsto \mathcal{L} \oplus \mathcal{O}_S^{\oplus(n-1)}.$$

This is injective, the determinant bundle is a left inverse. The $n = 1$ case then implies $|\mathcal{A}(n)| \geq \aleph_0$, we just need to bound this from above. This is immediate, though: using the same affine open cover $U, V \subseteq S$, note that any locally free $\mathcal{E} \in \mathfrak{Coh}(S)$ is trivialized on both U, V , since they are spectra of local rings. One then gets an injection

$$\mathcal{A}(n) \rightarrow \mathrm{GL}(n, \Gamma(U \cap V, \mathcal{O}_S)) \cong \mathrm{GL}(n, K),$$

by choosing representatives of classes of transition matrices. The codomain is countable because K is, and we're done. $\square$

What we have developed thus far almost allows us to characterize the topological spaces $\mathfrak{X}(S)$ for admissible S . We need to address one more technicality before handling these spaces.

Lemma 3: Let S be admissible and let $\mathcal{E} \in \mathfrak{Coh}(S)$ be locally free. The natural action of

$$\mathrm{Aut}_S\left(\underline{\mathrm{Spec}}_S(\mathrm{Sym}^\bullet(\mathcal{E}^\vee))\right)$$

on the set

$$\mathrm{hom}_{\mathrm{Sch}/S}(S, \underline{\mathrm{Spec}}_S(\mathrm{Sym}^\bullet(\mathcal{E}^\vee)))$$

is transitive.

Proof: Let $\varphi, \psi : S \rightarrow \underline{\mathrm{Spec}}_S(\mathrm{Sym}^\bullet(\mathcal{E}^\vee))$ be morphisms of S -schemes. By the natural isomorphism

$$\mathrm{hom}_{\mathrm{Sch}/S}(S, \underline{\mathrm{Spec}}_S(\mathrm{Sym}^\bullet(\mathcal{E}^\vee))) \cong H^0(S, \mathcal{E})$$

these maps correspond to sections $s_\varphi, s_\psi \in H^0(S, \mathcal{E}) \cong \mathrm{hom}_{\mathfrak{Coh}(S)}(\mathcal{O}_S, \mathcal{E})$. Define a natural transformation of functors $F_{\mathcal{E}}(-) \rightarrow F_{\mathcal{E}}(-)$ by declaring

$$\begin{aligned} F_{\mathcal{E}}(\phi : T \rightarrow S) &\cong H^0(T, \phi^* \mathcal{E}) \rightarrow H^0(T, \phi^* \mathcal{E}) \\ \alpha &\mapsto \alpha + \phi^* s_\psi - \phi^* s_\varphi. \end{aligned}$$

This is an automorphism of the functor $F_{\mathcal{E}}$, and so the Yoneda lemma implies there is an automorphism of S -schemes $\sigma : \underline{\mathrm{Spec}}_S(\mathrm{Sym}^\bullet(\mathcal{E}^\vee)) \rightarrow \underline{\mathrm{Spec}}_S(\mathrm{Sym}^\bullet(\mathcal{E}^\vee))$ inducing it.

Now compute the composition $\sigma \circ \varphi : S \rightarrow \underline{\mathrm{Spec}}_S(\mathrm{Sym}^\bullet(\mathcal{E}^\vee))$. In $H^0(S, \mathcal{E})$, this corresponds to the section

$$s_\varphi + s_\psi - s_\varphi = s_\psi,$$

hence we have $\sigma \circ \varphi = \psi$ and we're done. $\square$

The main reason we need *Lemma 3* is the following corollary.

Corollary 4: Let S be admissible and let $\mathcal{E}, \mathcal{F} \in \mathfrak{Coh}(S)$ be locally free. If we have

$$\underline{\mathrm{Spec}}_S(\mathrm{Sym}^\bullet(\mathcal{E}^\vee)) \cong \underline{\mathrm{Spec}}_S(\mathrm{Sym}^\bullet(\mathcal{F}^\vee))$$

as S -schemes, then $\mathcal{E} \cong \mathcal{F}$ as coherent sheaves.

Proof: Assume the premise. The functorial correspondence

$$\mathrm{hom}_{\mathrm{Sch}/S}(S, \underline{\mathrm{Spec}}_S(\mathrm{Sym}^\bullet(\mathcal{E}^\vee))) \cong H^0(S, \mathcal{E})$$

yields a canonical closed embedding $0_{\mathcal{E}} : S \rightarrow \underline{\mathrm{Spec}}_S(\mathrm{Sym}^\bullet(\mathcal{E}^\vee))$ corresponding to the trivial section $0 \in H^0(S, \mathcal{E})$, with a similar map $0_{\mathcal{F}}$ for $\mathcal{F}$. By *Lemma 3*, we may assume that the isomorphism

$$\psi : \underline{\mathrm{Spec}}_S(\mathrm{Sym}^\bullet(\mathcal{E}^\vee)) \xrightarrow{\cong} \underline{\mathrm{Spec}}_S(\mathrm{Sym}^\bullet(\mathcal{F}^\vee))$$

commutes with these maps. In that case, though, the conormal sheaves $\mathcal{C}_{\mathcal{E}} := \mathcal{I}_{0_{\mathcal{E}}}/\mathcal{I}_{0_{\mathcal{E}}}^2$ and $\mathcal{C}_{\mathcal{F}} := \mathcal{I}_{0_{\mathcal{F}}}/\mathcal{I}_{0_{\mathcal{F}}}^2$ corresponding to the zero sections satisfy

$$\psi^* \mathcal{C}_{\mathcal{F}} \cong \mathcal{C}_{\mathcal{E}}.$$

Pulling back along $0_{\mathcal{E}}$ and using $0_{\mathcal{E}}^* \psi^* = 0_{\mathcal{F}}^*$ then gives $\mathcal{E}^{\vee} \cong \mathcal{F}^{\vee}$, and then taking duals gives the result. $\square$

We are now, finally, in a position to handle the spaces $\mathfrak{X}(S)$. Up to homeomorphism, there are only two of them.

Lemma 5: Let D denote $\mathbb{N}$ with the discrete topology and let I denote $\mathbb{N}$ with the indiscrete topology. If S is admissible, then $\mathfrak{X}(S)$ is either homeomorphic to D or $\{*\} \sqcup (D \times I)$.

Proof: We split into two cases:

Case I: S is affine

If $\mathcal{E} \in \mathfrak{Coh}(S)$ is locally free, then the corresponding contravariant functor $F_{\mathcal{E}}(-)$ is representable by the relative spectrum

$$\underline{\mathrm{Spec}}_S(\mathrm{Sym}^{\bullet}(\mathcal{E}^{\vee})) \rightarrow S,$$

this is classical. Corollary 2 from the source material directly implies that this class of examples is the only one, that is, if $F_{\mathcal{E}} \in \mathfrak{X}(S)$, then $\mathcal{E}$ is locally free and the corresponding fine moduli scheme is of the above form.

We argue that in the affine case, the dimension map $\mathfrak{X}(S) \rightarrow \mathbb{Z}$ is injective, this will force $\mathfrak{X}(S) \cong D$. This essentially follows immediately from the fact that the only locally free sheaves on S in the affine case are trivial. Indeed, S is the spectrum of a semilocal Noetherian ring, as mentioned before, any locally free $\mathcal{E} \in \mathfrak{Coh}(S)$ is trivial. Writing $\mathcal{E} \cong \mathcal{O}_S^{\oplus \mathrm{rk}(\mathcal{E})}$, we get an isomorphism of S -schemes

$$\underline{\mathrm{Spec}}_S(\mathrm{Sym}^{\bullet}(\mathcal{E}^{\vee})) \cong \mathbb{A}_S^{\mathrm{rk}(\mathcal{E})}$$

and hence, if we write $\Lambda(S) = \{\mathbb{A}_S^n\}_{n=0}^{\infty}$, then the map $\mathfrak{X}(S) \rightarrow \mathbb{Z}$ factors as

$$\begin{array}{ccc} \mathfrak{X}(S) & \xrightarrow{\cong} & \Lambda(S) \\ & \searrow & \downarrow \mathrm{dim}(-) \\ & & \mathbb{Z} \end{array}$$

This concludes *Case I*.

Case II: S is not affine

In this case, we have $\mathfrak{X}(S) \cong \{*\} \bigsqcup (D \times I)$. To prove this, first note that by *Corollary 4*, if we have two rank $n > 0$ locally free sheaves $\mathcal{E}, \mathcal{F} \in \mathfrak{Coh}(S)$ with

$$\underline{\mathrm{Spec}}_S(\mathrm{Sym}^\bullet(\mathcal{E}^\vee)) \cong \underline{\mathrm{Spec}}_S(\mathrm{Sym}^\bullet(\mathcal{F}^\vee))$$

as S -schemes, then $\mathcal{E} \cong \mathcal{F}$. Now write $\varphi : \mathfrak{X}(S) \rightarrow \mathbb{Z}$ for the dimension map. Then since the codomain is discrete and the topology on $\mathfrak{X}(S)$ is induced by φ , the space in question is a disjoint union of fibers

$$\mathfrak{X}(S) \cong \bigsqcup_{n \in \mathbb{Z}} \varphi^{-1}(n).$$

The fibers $\varphi^{-1}(n)$ are either empty or homeomorphic to the indiscretely-topologized set of S -isomorphism classes of schemes of the form

$$\underline{\mathrm{Spec}}_S(\mathrm{Sym}^\bullet(\mathcal{E}^\vee))$$

for $\mathcal{E} \in \mathfrak{Coh}(S)$ locally free of rank $n - 1$, again by *Corollary 2* of the source material. Using the notation from *Lemma 2*, the map $\mathcal{A}(n - 1) \rightarrow \varphi^{-1}(n)$ given by

$$\mathcal{E} \mapsto \underline{\mathrm{Spec}}_S(\mathrm{Sym}^\bullet(\mathcal{E}^\vee))$$

is bijective, and *Lemma 2* then implies that

$$\mathfrak{X}(S) \cong \varphi^{-1}(1) \bigsqcup_{n \geq 2} I.$$

The fiber $\varphi^{-1}(1)$ is a point, containing only the functor

$$F_0(-) \cong \mathrm{hom}_{\mathrm{Sch}/S}(-, S),$$

so we get

$$\mathfrak{X}(S) \cong \{*\} \bigsqcup_{n \geq 2} I \cong \{*\} \bigsqcup (D \times I).$$

□

We can now conclude. We have

$$R \cong \mathbb{Q}^3.$$

To see this, let D, I be as in the statement of *Lemma 5*. Relabel $X \cong D$ and $Y \cong \{*\} \bigsqcup (D \times I)$, and define $Z := X \times Y$. Note, then, that in the semigroup $\mathcal{S}$ we have $X^2 = X$ and $Y^2 = Y$, with $XZ = YZ = Z^2 = Z$. The semigroup $\mathcal{S}$ then only has 3 elements.

The ring R is, as a $\mathbb{Q}$ -vector space, the direct sum

$$R \cong \mathbb{Q} \cdot X \oplus \mathbb{Q} \cdot Y \oplus \mathbb{Q} \cdot Z.$$

Now, notice that despite the fact that $\mathcal{S}$ is not unital, R actually is: we have $1_R = x + y - z$. Moreover, the following is a complete set of orthogonal idempotents in R :

$$\{x - z, y - z, z\}.$$

We get a direct product decomposition

$$R \cong \mathbb{Q}^3$$

and hence $\text{Spec}(R)$ has 3 connected components. $\square$

Domain: Algebraic Number Theory

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Prompt: If G is a finite group, then we say that a Galois extension K/F is a G -extension if $\text{Gal}(K/F) \cong G$. We let $g(F, G)$ denote the number of G -extensions of a field F , up to isomorphism. For an integer $n \geq 1$, let D_n denote the dihedral group of order $2n$ and Dic_n the dicyclic group of order $4n$. We have $g(\mathbb{Q}_5, D_5) = 3$, we have $g(\mathbb{Q}_p, D_6) = 3, 6$ as $p = 2, 3$, respectively, and we have $g(\mathbb{Q}_p, \text{Dic}_3) = 4, 2, 2$ as $p = 2, 3, 5$, respectively. We also have $g(\mathbb{Q}_2, S_4) = 3$. Recall that if $\mathbb{Q}_2^{\text{nil}}$ is the maximal pro-2 extension of $\mathbb{Q}_2$, then $\text{Gal}(\mathbb{Q}_2^{\text{nil}}/\mathbb{Q}_2)$ admits the presentation $\langle x, y, z \mid x^2 y^3 z^{-1} y z \rangle$ as a topological group.

For an integer $d \geq 1$, let $c(\mathbb{Q}_p, d)$ denote the number of isomorphism classes of non-Galois extensions $K/\mathbb{Q}_p$ of degree d . Let $0 \leq N < 10000$ be the unique integer such that

$$\sum_{p \in \{2, 3, 5\}} p \sum_{d=1}^{16} c(\mathbb{Q}_p, d) d \equiv N \pmod{10000}.$$

What is N ?

Explanation: Table 1.1 gives the number of isomorphism classes of degree- d extensions of $\mathbb{Q}_p$ for each $1 \leq d \leq 16$ and $p = 2, 3, 5$, so we count Galois extensions and subtract. For topological groups G, H we let $\Phi(G, H)$ be the number of continuous homomorphisms $G \rightarrow H$ and $\Phi_0(G, H)$ the number of such homomorphisms which are surjective. We need only consider $d \geq 3$.

We begin with the abelian extensions, where we can use local class field theory. Fix a finite abelian group G of order d and a prime p . We will not consider the case $p = 2$ and $d = 16$, as the paper allows us to circumvent direct counting in this case. The number of G -extensions of $\mathbb{Q}_p$ is

$$\frac{|\Phi_0(\mathbb{Q}_p^\times, G)|}{|\text{Aut}(G)|} = \frac{|\Phi_0(\mathbb{Z} \times \mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}_p, G)|}{|\text{Aut}(G)|}.$$

It follows that $g(\mathbb{Q}_p, G) = 0$ if G requires more than 3 elements to generate, and $g(\mathbb{Q}_p, G) \leq 1$ when d is prime to p and $p - 1$, with equality if and only if G is cyclic.

We have $\text{Aut}(\mathbb{Z}/d\mathbb{Z}) = \varphi(d)$ and $\Phi(\mathbb{Q}_p^\times, \mathbb{Z}/d\mathbb{Z}) = d \cdot \gcd(p-1, d) p^{\nu_p(d)}$. Now observe that

$$\Phi(\mathbb{Q}_p^\times, \mathbb{Z}/d\mathbb{Z}) = \sum_{b|d} \Phi_0(\mathbb{Q}_p^\times, \mathbb{Z}/b\mathbb{Z}),$$

so by Möbius inversion,

$$\Phi_0(\mathbb{Q}_p^\times, \mathbb{Z}/d\mathbb{Z}) = \sum_{ab=d} \mu(a) \Phi(\mathbb{Q}_p^\times, \mathbb{Z}/b\mathbb{Z}) = \sum_{ab=d} \mu(a) b \cdot \gcd(p-1, b) p^{\nu_p(b)}.$$

We start with the non-cyclic cases. If $G \cong (\mathbb{Z}/2\mathbb{Z})^r$, then any G -extension can be written as a composite of r quadratic extensions in exactly $2^r - 1$ distinct ways, thus

$$g(\mathbb{Q}_p, (\mathbb{Z}/2\mathbb{Z})^r) = \binom{g(\mathbb{Q}_p, \mathbb{Z}/2\mathbb{Z})}{r} \cdot \frac{1}{2^r - 1}.$$

We obtain that

$$g(\mathbb{Q}_p, (\mathbb{Z}/2\mathbb{Z})^r) = 7, 1, 1, 0, 1, 0 \quad \text{as} \quad (p, r) = (2, 2), (2, 3), (3, 2), (3, 3), (5, 2), (5, 3).$$

If $G \cong \mathbb{Z}/2^j\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ with $j \geq 2$, then G can be written as a composite of cyclic extensions of degree 2^j and 2 in four distinct ways, thus

$$g(\mathbb{Q}_p, G) = \frac{g(\mathbb{Q}_p, \mathbb{Z}/2^j\mathbb{Z})(g(\mathbb{Q}_p, \mathbb{Z}/2\mathbb{Z}) - 1)}{4}.$$

We obtain that

$$\begin{aligned} g(\mathbb{Q}_p, \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}) &= 18, 1, 3 & \text{as} & \quad p = 2, 3, 5 \\ g(\mathbb{Q}_p, \mathbb{Z}/8\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}) &= 36, 1, 2 & \text{as} & \quad p = 2, 3, 5. \end{aligned}$$

It is clear that $g(\mathbb{Q}_p, \mathbb{Z}/4\mathbb{Z} \times (\mathbb{Z}/2\mathbb{Z})^2) = 0$ for $p = 3, 5$, and likewise that $g(\mathbb{Q}_p, (\mathbb{Z}/3\mathbb{Z})^2) = 0, 1, 0$ as $p = 2, 3, 5$. Finally, we have

$$g(\mathbb{Q}_p, \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}) = g(\mathbb{Q}_p, \mathbb{Z}/3\mathbb{Z})g(\mathbb{Q}_p, (\mathbb{Z}/2\mathbb{Z})^2)$$

, thus

$$g(\mathbb{Q}_p, \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}) = 7, 4, 1 \quad \text{as} \quad p = 2, 3, 5.$$

Using our formula to count cyclic extensions and adding in the non-cyclic abelian extensions, we summarize the number of abelian extensions in the following table:

	d=3	4	5	6	7	8	9	10	11	12	13	14	15	16
p=2	1	19	1	7	1	43	1	7	1	19	1	7	1	
p=3	4	3	1	12	1	3	13	3	1	12	1	3	4	3
p=5	1	7	6	3	1	7	1	18	1	7	1	3	6	7

Now we consider non-abelian groups G of order d . If p does not divide d , then any G -extension of $\mathbb{Q}_p$ is tamely ramified, and so the number of such extensions is $\mathbb{Q}_p$ is

$$\frac{|\Phi_0(G_p^{\text{tr}}, G)|}{|\text{Aut}(G)|} = \frac{|\Phi_0(\langle \sigma, \tau \mid \tau \sigma \tau^{-1} = \sigma^p \rangle, G)|}{|\text{Aut}(G)|},$$

where G_p^{tr} is the Galois group of the maximal tamely ramified extension of $\mathbb{Q}_p$. Below we assume that K is a G -extension of $\mathbb{Q}_p$. We let $G_0 \supseteq G_1$ denote the inertia and wild inertia subgroups. Recall that G/G_0 is cyclic, G_0 is a p -group, and G_1/G_0 is cyclic of order prime to p .

Suppose that $G \cong S_3$. Then G -extensions are in bijection with non-Galois degree-3 extensions, so we simply use the counts from Table 1.1 and subtract $g(\mathbb{Q}_p, \mathbb{Z}/3\mathbb{Z})$. We obtain that $g(\mathbb{Q}_p, S_3) = 1, 6, 1$ as $p = 2, 3, 5$.

Suppose that $G \cong D_4$. The presentation of G_p^{tr} shows that if p is odd, then necessarily $p \equiv 3 \pmod{4}$. In this case, g, h generate G and satisfy $hgh^{-1} = g^p$ if and only if g has order 4 and h is a non-central element of order 2, and thus there are 8 such pairs. Since $\text{Aut}(G) \cong G$, we see that there is a unique such extension. Then $g(\mathbb{Q}_p, D_4) = 1, 0$ as $p = 3, 5$. If $p = 2$, then we use the presentation $\langle x, y, z \mid x^2y^3z^{-1}yz \rangle$ of $\text{Gal}(\mathbb{Q}_2^{\text{nil}}/\mathbb{Q}_2)$. Write $G = \langle a, r \mid a^4, r^2, r a r a \rangle$. We have $g^2 = g^{-2} = e, a^2$ and $g^3 = g^{-1}$ for all $g \in G$. If $x^2 = e$, then y, z commute. Let $H = \langle y, z \rangle$. If H has order 2, then $\{y, z\}$ is equal to $\{g\}$ or $\{g, e\}$ for some order 2 element g . In either case, x can and must be any element of order 2 which does not commute with g . There are four pairs of non-commuting elements of G , yielding $2 \cdot 4 + 2 \cdot 2 \cdot 4 = 24$ valid triplets (x, y, z) . There are $4^2 - 4 = 12$ ways to choose y, z so that $H \cong \mathbb{Z}/4\mathbb{Z}$. In this case, x can and must be any element of $G \setminus H$, so there are 4 choices, yielding $12 \cdot 4 = 48$ triplets. There are $2 \cdot 3 \cdot 2 = 12$ ways to choose y, z so that $H \cong (\mathbb{Z}/2\mathbb{Z})^2$. In this case, x can and must be one of the two order 2 elements in $G \setminus H$, so we get $12 \cdot 2 = 24$ triplets. There are thus 96 triplets where $x^2 = e$. If $x = a^2$, then $x = a, a^3$ and y, z can and must be any pair of non-commuting elements. There are 12 such pairs, so we get $2 \cdot 2 \cdot 12 = 48$ triplets. There are thus 144 triplets total, so $g(\mathbb{Q}_2, D_4) = 18$.

The case $G \cong Q_8$ is similar. We have $\text{Aut}(G) \cong S_4$, and tame ramification again gives $g(\mathbb{Q}_p, Q_8) = 1, 0$ as $p = 3, 5$. When $p = 2$, we again count triplets (x, y, z) generating G such that $x^2y^{-1}z^{-1}yz$ is trivial. If $x^2 = e$, then $x \in Z(G)$ while y, z commute, a contradiction. Then (x, y, z) is a valid triplet if and only if x has order 4 and y, z do not commute. There are thus $6 \cdot 6 \cdot 4 = 144$ triplets and $g(\mathbb{Q}_2, Q_8) = 6$.

Suppose that $G \cong D_5$. Since G/G_0 is cyclic, either $G_0 = G$ or G_0 has order 5. Since G_1/G_0 is cyclic of order prime to p and G_1 is a p -group, $G_0 = G$ forces $p = 5$. If $p = 2, 3$, then tame ramification shows $g(\mathbb{Q}_p, D_5) = 0$. We are given $g(\mathbb{Q}_5, D_5) = 3$.

Suppose that $G \cong D_6$. We are given $g(\mathbb{Q}_p, D_6) = 3, 6$ as $p = 2, 3$. If $p = 5$, tame ramification gives $g(\mathbb{Q}_5, D_6) = 1$.

Now suppose that $G \cong A_4$. Since G/G_0 is cyclic, it must be that G_0 is the 2-Sylow subgroup or $G_0 = G$. Since G_0/G_1 is cyclic of order prime to p and G_1 is a p -group, we see that necessarily $p = 2$. Table 1.1 gives 59 degree-4 extensions of $\mathbb{Q}_2$, and we counted 19 abelian extensions, so there are 40 non-abelian ones. Since $g(\mathbb{Q}_2, D_4) = 18$, exactly 36 of these have Galois group D_4 . The remaining 4 have Galois group A_4 or S_4 . We are given $g(\mathbb{Q}_2, S_4) = 3$, so it follows that $g(\mathbb{Q}_2, A_4) = 1$.

Suppose that $G \cong D_7$. Since G/G_0 is cyclic, either $G_0 = G$ or G_0 has order 7. Since G_1/G_0 is cyclic of order prime to p and G_1 is a p -group, the former case forces $p = 7$. Tame ramification gives $g(\mathbb{Q}_p, D_7) = 0$ for $p = 2, 3, 5$.

Now suppose that G is non-abelian of order 16. If $p = 2$, we do not need to count; the beginning of Section 5.5 of the paper directly gives that $c(\mathbb{Q}_2, 16) =$

251. In the remaining cases, we have tame ramification. The group G must be generated by two elements g, h satisfying $hgh^{-1} = g^p$. The only such groups are generalized quaternion group, and semidirect products of the form $\mathbb{Z}/8\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$ or $\mathbb{Z}/4\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z}$. Since we are not considering any primes that are congruent to 7 modulo 8, we need not consider the generalized quaternion or dihedral group are ruled out. Then for each $p = 3, 5$ we get semidirect products $\mathbb{Z}/8\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$ and for $p = 3$ we also get $\mathbb{Z}/4\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z}$. For pair of a prime $p = 3, 5$ and a valid group G , we find pairs x, y generating G such that $xyx^{-1} = x^p$. In particular, counting such pairs where the orders of x, y are the same as the orders of the respective generators of the semidirect product allows us to count automorphisms.

Let $p = 5$ and $G \cong \langle a, b \mid a^8, b^2, baba^{-5} \rangle$. In this case x must have order 8. In any case, y can and must be any of the 8 elements in $G \setminus \langle x \rangle$, yielding $8 \cdot 8$ homomorphisms. There are 2 choices of y with order 2, yielding $8 \cdot 2 = 16$ automorphisms. Therefore $g(\mathbb{Q}_5, G) = 4$.

Let $p = 3$. First consider $G \cong \langle a, b \mid a^8, b^2, baba^{-3} \rangle$. The order of x must be 4 or 8. In the former case, the order of y must be 8 since all order 4 subgroups of G contain a^4 . But then $x = a^j b, y = a^k$ with j, k odd and $xyx^{-1} = a^{j-2i} b \neq a^j b$ unless $i = 0$, contradicting that x, y generate G . Therefore x has order 8 and so y can have order 2 or order 4. The element y can and must be any of the 4 non-central element of order 2 or any of the 4 order 4 elements not contained in $\langle a \rangle$, giving $4 \cdot 4$ automorphisms and $4(4 + 4) = 32$ surjective homomorphisms. Then $g(\mathbb{Q}_3, G) = 2$. Now consider $G \cong \langle a, b \mid a^4, b^2, bab^{-1}a \rangle$. The only order 2 elements are central, and thus x, y both have order 4. It thus follows that $g(\mathbb{Q}_3, G) = 1$. Finally, consider $G \cong \langle a, b \mid a^8, b^2, baba^{-5} \rangle$ again. There are 16 generating pairs (x, y) with $xyx^{-1} = x^3$, and we already know $|\text{Aut}(G)| = 16$, so $g(\mathbb{Q}_3, G) = 1$. For other choices of p and G we have $g(\mathbb{Q}_p, G) = 0$.

Adding the counts for abelian and non-abelian Galois extensions, we summarize the number of Galois extensions in the table below:

	d=3	4	5	6	7	8	9	10	11	12	13	14	15	16
p=2	1	19	1	8	1	67	1	7	1	27	1	7	1	251
p=3	4	3	1	18	1	5	13	3	1	20	1	3	4	7
p=5	1	7	6	4	1	7	1	21	1	10	1	3	6	11

Subtracting these values from the corresponding values in Table 1.1, we obtain the following table giving values of $c(\mathbb{Q}_p, d)$:

	d=3	4	5	6	7	8	9	10	11	12	13	14	15	16
p=2	1	40	1	39	1	1756	2	151	1	5466	1	583	3	889860
p=3	6	2	1	57	1	3	782	3	1	765	1	3	1168	6
p=5	1	0	20	3	1	4	2	237	1	7	1	3	1006	6

We find that

$$\sum_{p \in \{2, 3, 5\}} p \sum_{d=1}^{16} c(\mathbb{Q}_p, d) d = 28847558.$$

Modulo 10000, this is 7558.

Domain: Representation Theory

Date: May 24, 2026

Paper: <https://arxiv.org/pdf/1304.0754>

Prompt: Let A be an Artin R -algebra with R an Artinian ring. We denote by $\mathcal{C}_A$ the abelian category of all R -functors $F : \text{mod}(A) \rightarrow \text{mod}(R)$.

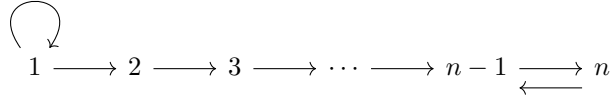
For a module M , we denote by $\text{add}(M)$ the subcategory of all modules that are direct sums of direct summands of M .

Definition: A pair (X, Y) of objects in $\text{add}(M)$ is called a d -Division of M with $d \in \mathbb{Z}_{>0}$ if the following three conditions hold:

- (a) $\text{add}(X) \cap \text{add}(Y) = \{0\}$,
- (b) $\text{Ext}_A^d(X, -) \not\cong \text{Ext}_A^d(Y, -)$ in $\mathcal{C}_A$,
- (b) $\text{Ext}_A^{d+1}(X, -) \cong \text{Ext}_A^{d+1}(Y, -)$ in $\mathcal{C}_A$.

For any module M , define $\text{div}(M) = \max\{d : \text{there is a } d\text{-Division of } M\}$. If no d -Division exists, we set $\text{div}(M) = 0$.

Consider the bound quiver algebra $B_n = \frac{kQ}{J^3}$, where k is an algebraically closed field and Q is the quiver



with $n \geq 3$ and J is the Jacobson radical.

Denote by S_i the simple module associated to vertex i in Q .

When $n = 2026$, what is the value of $\text{div}(\oplus_{i=1}^n S_i)$?

Explanation: We do the calculations for any n and then substitute $n = 2026$. We start by describing the indecomposable projective B_n -modules using their Loewy sequences. We denote by P_i the indecomposable projective associated to vertex i .

- $P_1 = \begin{array}{c} 1 \\ 1 \ 2 \\ 1 \ 2 \ 3 \end{array}$
- $P_2 = \begin{array}{c} 2 \\ 3 \\ 4 \end{array}$
- $\vdots$
- $P_{n-2} = \begin{array}{c} n-2 \\ n-1 \\ n \end{array}$

$$\bullet P_{n-1} = \begin{array}{c} n-1 \\ n \\ n-1 \end{array}$$

$$\bullet P_n = \begin{array}{c} n \\ n-1 \\ n \end{array}$$

The syzygies of the simple modules are as follows:

$$\bullet \Omega(S_1) = \begin{array}{c} 1 \\ 1 \end{array} \oplus \begin{array}{c} 2 \\ 2 \end{array} \oplus \begin{array}{c} 3 \\ 3 \end{array}$$

$$\bullet \Omega^2(S_1) = S_1 \oplus S_2 \oplus S_3 \oplus S_4$$

$$\bullet \Omega(S_2) = \begin{array}{c} 3 \\ 4 \end{array}$$

$\vdots$

$$\bullet \Omega(S_{n-2}) = \begin{array}{c} n-1 \\ n \end{array}$$

$$\bullet \Omega(S_{n-1}) = \begin{array}{c} n \\ n-1 \end{array}$$

$$\bullet \Omega(S_n) = \begin{array}{c} n-1 \\ n \end{array}$$

Note that if n is small, say $n = 3$, the above formulas have to be adjusted accordingly. For instance P_2 won't be $\begin{array}{c} 2 \\ 3 \end{array}$ because there is no vertex 4. Since

$$2 = 3 - 1 = n - 1, \text{ then } P_2 = P_{n-1} = \begin{array}{c} n-1 \\ n \\ n-1 \end{array}, \text{ that is } P_2 = \begin{array}{c} 2 \\ 3 \end{array} \text{ if } n = 3. \text{ The same}$$

applies to the syzygy calculations.

In order to find $\text{div}(\oplus_{i=1}^n S_i)$ we need to detect d -Divisions. We recall an elementary equivalence saying that $\text{Ext}_{B_n}^1(M, -) \cong \text{Ext}_{B_n}^1(N, -)$ in $\mathcal{C}_{B_n}$ if and only if M, N differ by a projective summand, that is there are P, Q projective such that $M \oplus P \cong N \oplus Q$ in $\text{mod } B_n$. This result is part of Proposition 3.1 in the paper.

A direct consequence of the above is that $\text{Ext}_{B_n}^i(M, -) \cong \text{Ext}_{B_n}^i(N, -)$ in $\mathcal{C}_{B_n}$ for all $i \geq m + 1$ if and only if there are projectives P', Q' such that $\Omega^m M \oplus P' \cong \Omega^m N \oplus Q'$. This result is part of Corollary 3.3 in the paper.

Now we observe that there are no simple projective B_n -modules and no syzygy of a simple module has a projective summand. This means that d -Divisions of $S_1 \oplus \cdots \oplus S_n$, if they exist, must come from pairs of summands S, S' for which $\Omega^m S \cong \Omega^m S'$ for some $m \geq 1$.

Let us write down the set of isoclasses of indecomposable summands of $S_1 \oplus \dots \oplus S_n$. In each step we will substitute each summand by the isoclass of its syzygy and check whether there are two or more that are equal.

Step 0: $\{S_1, \dots, S_{n-6}, S_{n-5}, S_{n-4}, S_{n-3}, S_{n-2}, S_{n-1}, S_n\}$

Step 1: $\{\Omega(S_1), \dots, \begin{smallmatrix} n-5 & n-4 & n-3 & n-2 & n-1 & n & n-1 \\ n-4' & n-3' & n-2' & n-1' & n & n-1' & n \end{smallmatrix}\}$

Here we detect one pair of isomorphic terms, that is $\Omega(S_{n-2}) \cong \Omega(S_n) \cong \begin{smallmatrix} n-1 \\ n \end{smallmatrix}$, which gives a 1-Division. We can set one of them as zero, so we don't count that isomorphism again, for instance we will set the last element to zero. Let us continue with the next step:

Step 2: $\{\Omega^2 S_1, \dots, S_{n-3}, S_{n-2}, S_{n-1}, S_n, S_{n-1}, S_n, 0\}$

Here we detect two pairs of isomorphisms: $\Omega^2(S_{n-3}) \cong \Omega^2(S_{n-1})$ and $\Omega^2(S_{n-4}) \cong \Omega^2(S_{n-2})$. This produces a 2-Division. Note that, for any $j \geq 0$, $\Omega^j(S_1)$ is not isomorphic to $\Omega^j S'$ for any S' summand of $S_2 \oplus \dots \oplus S_n$, so we write $\Omega^j(S_1)$ instead of its indecomposable decomposition, which is large and irrelevant for this calculation.

If we continue from here, we see that on each step we produce a new division and the number of elements in the corresponding set decreases in one or two since we are putting zeros to indicate where a division has been found. Note also that this process will result in one of the following two possibilities:

- The set reduces to $\{\Omega^m(S_1), \begin{smallmatrix} n \\ n-1 \end{smallmatrix}, \begin{smallmatrix} n-1 \\ n \end{smallmatrix}, 0, \dots, 0\}$ when $m = \text{div}(\oplus_{i=1}^n S_i)$.
- The set reduces to $\{\Omega^m(S_1), S_{n-1}, S_n, 0, \dots, 0\}$ when $m = \text{div}(\oplus_{i=1}^n S_i)$.

Therefore, all we need to do is, given n , find the step in which we arrive at one of the sets above. We start with n elements and put aside three which will remain stable at the end. For the remaining $n-3$ we can eliminate three at a time every two steps, and when we can't eliminate three, we eliminate one at a time until we get to zero.

By integer division, there are unique integers q, r such that $n-3 = q \cdot 3 + r$, with $q \geq 0$ and $0 \leq r \leq 2$. Then

$$\text{div}(\oplus_{i=1}^n S_i) = 2q + r.$$

If $n = 3$, it is clear that $\text{div}(\oplus_{i=1}^n S_i) = 0$ because there are no d -divisions. As we can see the formula also gives $n-3 = 0 = 0 \cdot 3 + 0$, so $q = 0$ and $r = 0$.

Alternatively, we can use Theorem 3.6 in the paper and calculate $\phi(\oplus_{i=1}^n S_i)$ instead, where ϕ is the Igusa-Todorov function.

If $n = 2026$, then $n-3 = 2023 = 674 \cdot 3 + 1$. This means that we can eliminate three elements at a time for a total of 674 times. Each of those correspond to two

steps, so we will have a 1348-Division. At this point, we can do one more step and eliminate one more element before getting to a stable set. That last step corresponds to a 1349-Division and this is the maximum number of Divisions, since the set is stable from this step onward. Hence,

$$\text{div}(\oplus_{i=1}^n S_i) = 2q + r = 2 \cdot 674 + 1 = 1349.$$

Domain: Computational Group Theory (Tools Required)

Date: April 13, 2026

Paper: <https://arxiv.org/pdf/2405.15300>

Prompt: Let G be an almost simple group with socle $T = A_n$. A core-free subgroup $H \leq G$ is primitive if $H \cap T$ acts primitively on $\{1, \dots, n\}$. The primitive regularity number $R_{\text{prim}}(G)$ is the smallest positive integer k such that every k -tuple $(H_1, \dots, H_k)$ of core-free primitive subgroups of G is regular, i.e. such that there exist $g_1, \dots, g_k \in G$ with $\bigcap_{i=1}^k H_i^{g_i} = 1$. If G has no core-free primitive subgroup, $R_{\text{prim}}(G)$ is undefined. Suppose that G has a core-free primitive subgroup and that $R_{\text{prim}}(G) = 6$. Let m be the number of simple $GF(5)G$ -modules.

For a prime p , let n_p denote the number of p -blocks of G over a splitting p -modular system. What is $n_2 + n_3 + n_5 + n_7 + n_{11} + n_{13} + m$?

Explanation: The paper forces $G = A_8$, so it remains to compute $n_2, n_3, n_5, n_7, n_{11}, n_{13}$ and the number m of simple $GF(5)A_8$ -modules. For the block counts, I enumerated all partitions λ of 8, computed their p -cores, and then used the standard block parametrisation for A_n . For odd p , a pair of non self-conjugate p -cores contributes one p -block of A_8 , a self-conjugate core of positive weight contributes one block, and a self-conjugate core of weight 0 contributes two defect zero blocks. For $p = 2$, only weight 0 blocks split on passing from S_8 to A_8 , so each 2-core of positive weight contributes one block. This gives $(n_2, n_3, n_5, n_7, n_{11}, n_{13}) = (2, 4, 7, 10, 14, 14)$.

For m , one has to count simple modules over $GF(5)$, not over a splitting field. Thus m is the number of 5-regular $GF(5)$ -conjugacy classes of A_8 . I computed this from the 5-regular classes of A_8 by checking when a split class remains split under Frobenius $x \mapsto x^5$. The only relevant split 5-regular class is cycle type $(7, 1)$, and its two A_8 -classes fuse over $GF(5)$ because 5 is a nonsquare modulo 7, so this contributes 1 simple module rather than 2. Hence $m = 10$, and therefore $n_2 + n_3 + n_5 + n_7 + n_{11} + n_{13} + m = 2 + 4 + 7 + 10 + 14 + 14 + 10 = 61$.

```
from math import prod

def parts(n, cap=None):
    if n == 0:
        return []
    if cap is None or cap > n:
        cap = n
    out = []
    for a in range(cap, 0, -1):
        out += [(a,) + r for r in parts(n - a, a)]
    return out
```

```

def conj(lam):
    if not lam:
        return ()
    return tuple(sum(x >= j for x in lam) for j in range(1, max(lam) + 1))

def core(lam, p):
    r = len(lam) + 20
    S = {lam[i] + r - i - 1 for i in range(len(lam))}
    S |= {r - i - 1 for i in range(len(lam), r)}
    changed = True
    while changed:
        changed = False
        for b in sorted(list(S)):
            while b >= p and b - p not in S:
                S.remove(b)
                S.add(b - p)
                b -= p
            changed = True
    beta = sorted(S, reverse=True)
    return tuple(b - r + i for i, b in enumerate(beta, 1) if b - r + i > 0)

def oddblocks(p):
    ans = 0
    used = set()
    for g in sorted({core(lam, p) for lam in P}):
        if g in used:
            continue
        w = (8 - sum(g)) // p
        if g != conj(g):
            ans += 1
            used |= {g, conj(g)}
        elif w > 0:
            ans += 1
            used.add(g)
        else:
            ans += 2
            used.add(g)
    return ans

def splitA(lam):
    return all(x % 2 == 1 for x in lam) and len(set(lam)) == len(lam)

def jacobi(a, n):
    a %= n
    out = 1
    while a:

```

```

        while a % 2 == 0:
            a //= 2
            if n % 8 in (3, 5):
                out = -out
        a, n = n, a
        if a % 4 == 3 and n % 4 == 3:
            out = -out
        a %= n
    return out if n == 1 else 0

def gf5_simple_A8():
    ans = 0
    for lam in P:
        if (8 - len(lam)) % 2 != 0:
            continue
        if any(x % 5 == 0 for x in lam):
            continue
        if not splitA(lam):
            ans += 1
        else:
            ans += 2 if jacobi(5, prod(lam)) == 1 else 1
    return ans

P = parts(8)

n2 = sum(1 if (8 - sum(g)) // 2 > 0 else 2 for g in {core(lam, 2) for lam in P})
vals = (n2, oddblocks(3), oddblocks(5), oddblocks(7), oddblocks(11), oddblocks(13), gf5_simple_A8())
print(vals, sum(vals))

```

Domain: Combinatorics (Tools Required)

Date: May 11, 2026

Paper: <https://arxiv.org/pdf/2602.12130>

Prompt: A pattern is a nonempty finite sequence $\rho = (\rho_1, \dots, \rho_k)$ of non-negative integers whose set of values is exactly

$$\{0, 1, \dots, \max(\rho)\}.$$

For example, 021, 010, and 1302 are patterns, while 202 is not a pattern. An inversion sequence is a finite sequence $\sigma = (\sigma_1, \dots, \sigma_n)$ of nonnegative integers such that

$$0 \leq \sigma_i \leq i - 1 \text{ for every } i = 1, \dots, n.$$

For any finite sequence $a = (a_1, \dots, a_t)$, define its reduction $\text{red}(a)$ by replacing the smallest value appearing in a by 0, the next smallest distinct value by 1, and so on. For example,

$$\text{red}(4, 9, 4, 7) = (0, 2, 0, 1).$$

A sequence σ contains a pattern ρ if there is a subsequence

$$(\sigma_{i_1}, \dots, \sigma_{i_k}) \quad (1 \leq i_1 < \dots < i_k \leq n)$$

whose reduction is exactly ρ . Otherwise, σ avoids ρ .

A sequence τ is called a ρ minimal inversion sequence if all three of the following hold: τ is both a pattern and an inversion sequence, τ contains ρ , and there is no sequence $\nu \neq \tau$ such that ν is both a pattern and an inversion sequence, ν contains ρ , and τ contains ν .

For a fixed pattern ρ , let $c_n(\rho)$ be the number of ρ -minimal inversion sequences of length n . Define the profile of ρ to be the tuple

$$P(\rho) = (c_L(\rho), c_{L+1}(\rho), \dots, c_U(\rho)),$$

where L is the smallest n with $c_n(\rho) > 0$ and U is the largest n with $c_n(\rho) > 0$.

What is the number

$$N = \#\{P(\rho) : \rho \text{ is a pattern of length at most } 6\}?$$

Explanation: For the length of at most 5, Table 2 lists 27 profiles. For the next maximum length of 6 we use computation. The result of running

```
python3 minimal_inversion_profiles.py 6 --processes 4 --show-profiles
```

is 79. The run takes a couple of minutes, but not excessively long and model correct runs with the paper show that this is doable for the model in the given environment within the timeouts set for tool use.

```

from __future__ import annotations
import argparse
import itertools
import math
import multiprocessing as mp
import os
import sys
import time
from collections import Counter
from functools import lru_cache
from typing import Dict, Iterable, Iterator, List, Sequence, Tuple

Pattern = Tuple[int, ...]

def reduction(seq: Sequence[int]) -> Pattern:
    vals = sorted(set(seq))
    rank = {v: i for i, v in enumerate(vals)}
    return tuple(rank[x] for x in seq)

def mdd(seq: Sequence[int]) -> int:
    # mdd(seq) = max(seq_i - i + 1) (1 based index)
    # in 0 based indexing this is max( seq[i] - i)
    return max(v - i for i, v in enumerate(seq))

def sat_positions(seq: Sequence[int]) -> List[int]:
    m = mdd(seq)
    return [i for i, v in enumerate(seq) if v - i == m]

@lru_cache(maxsize=None)
def rho_relations(rho: Pattern) -> Tuple[Tuple[Tuple[int, ...], Tuple[int, ...], Tuple[int, ...]], Tuple[int, ...], Tuple[int, ...]]:
    """
    For each position j in rho store which earlier positions p satisfy
    rho[p] < rho[j], rho[p] == rho[j], rho[p] > rho[j].
    This will lets us test occurrences without recomputing reductions.
    """
    rel = []
    for j in range(len(rho)):
        lt, eq, gt = [], [], []
        for p in range(j):
            if rho[p] < rho[j]:
                lt.append(p)
            elif rho[p] == rho[j]:
                eq.append(p)
            else:
                gt.append(p)

```



```

        else:
            gt.append(p)
            rel.append((tuple(lt), tuple(eq), tuple(gt)))
    return tuple(rel)

@lru_cache(maxsize=None)
def occurrence_index_tuples(sigma: Pattern, rho: Pattern) -> Tuple[Tuple[int, ...], ...]:
    """ All occurrences of rho in sigma """
    n, k = len(sigma), len(rho)
    rel = rho_relations(rho)
    chosen_idx = [0] * k
    chosen_val = [0] * k
    out: List[Tuple[int, ...]] = []

    def rec(j: int, start: int) -> None:
        if j == k:
            out.append(tuple(chosen_idx))
            return
        max_i = n - (k - j) + 1
        lt, eq, gt = rel[j]
        for i in range(start, max_i):
            v = sigma[i]
            ok = True
            for p in lt:
                if not (chosen_val[p] < v):
                    ok = False
                    break
            if ok:
                for p in eq:
                    if not (chosen_val[p] == v):
                        ok = False
                        break
            if ok:
                for p in gt:
                    if not (chosen_val[p] > v):
                        ok = False
                        break
            if ok:
                chosen_idx[j] = i
                chosen_val[j] = v
                rec(j + 1, i + 1)

    rec(0, 0)
    return tuple(out)

```

```

@lru_cache(maxsize=None)
def is_rho_minimal(sigma: Pattern, rho: Pattern) -> bool:
    """
    rho minimality using Prop 4 from the paper
    Assume here sigma is alr both a pattern and an inversion sequence
    """
    occs = occurrence_index_tuples(sigma, rho)
    if not occs:
        return False

    counts = Counter(sigma)
    last_sat = max(i for i, v in enumerate(sigma) if v == i) # sigma is an inversion sequence
    required_tail = set(range(last_sat, len(sigma)))

    for occ in occs:
        occ_set = set(occ)
        if not required_tail.issubset(occ_set):
            return False

        occ_vals = {sigma[i] for i in occ}
        for val, cnt in counts.items():
            if val not in occ_vals and cnt < 2:
                return False

    return True

def gen_rgs(n: int) -> Iterator[Pattern]:
    """
    Generate RGS of length n
    We get every pattern from a unique RGS by permuting labels
    """
    if n <= 0:
        return
    arr = [0] * n

    def rec(i: int, mx: int) -> Iterator[Pattern]:
        if i == n:
            yield tuple(arr)
            return
        for x in range(mx + 2):
            arr[i] = x
            yield from rec(i + 1, max(mx, x))

    yield from rec(1, 0)

```

```

def gen_patterns_of_length(n: int) -> Iterator[Pattern]:
    for rgs in gen_rgs(n):
        m = max(rgs)
        for perm in itertools.permutations(range(m + 1)):
            yield tuple(perm[x] for x in rgs)

def weak_compositions(total: int, parts: int) -> Iterator[Tuple[int, ...]]:
    if parts == 0:
        if total == 0:
            yield ()
        return

    comp = [0] * parts

    def rec(i: int, rem: int) -> Iterator[Tuple[int, ...]]:
        if i == parts - 1:
            comp[i] = rem
            yield tuple(comp)
            return
        for x in range(rem + 1):
            comp[i] = x
            yield from rec(i + 1, rem - x)

    yield from rec(0, total)

def filler_assignments(slot_positions: Sequence[int], counts_dict: Dict[int, int]) -> Iterator[Dict[int, int]]:
    """
    Generate all multiset assignments to given slot positions
    A value x can be placed in slot position p iff x <= p as this is inversion sequence constraints
    """
    values = [v for v, c in counts_dict.items() if c > 0]
    counts = [counts_dict[v] for v in values]
    max_allowed = list(slot_positions)
    arr = [None] * len(slot_positions)

    # how many remaining slots can still hold value v
    canfit = {v: [0] * (len(slot_positions) + 1) for v in values}
    for v in values:
        cnt = 0
        for i in range(len(slot_positions) - 1, -1, -1):
            if max_allowed[i] >= v:
                cnt += 1

```

```

        canfit[v][i] = cnt

def rec(i: int) -> Iterator[Tuple[int, ...]]:
    if i == len(slot_positions):
        yield tuple(arr)
        return

    for vi, v in enumerate(values):
        if counts[vi] > canfit[v][i]:
            return

    choices = [vi for vi, v in enumerate(values) if counts[vi] and v <= max_allowed[i]]
    for vi in choices:
        counts[vi] -= 1
        arr[i] = values[vi]
        yield from rec(i + 1)
        counts[vi] += 1

yield from rec(0)

def generate_rho_minimals( rho: Pattern) -> List[Pattern]:
    """
    Generate all rho minimal inversion sequences using
    Sec 5.2 of the paper
    then confirm each candidate with Prop 4
    """
    k = len(rho)
    m = mdd(rho)

    if m == 0:
        return [rho]

    d = max(rho) + 1
    s0 = min(sat_positions(rho)) # 0? index of the leftmost saturated entry
    out: set[Pattern] = set()

    for t in range(m, 2 * m + 1):
        v = t - m #number of new values introduced
        occ_s_val = rho[s0] + v
        total_distinct = d + v

        # New values all lie strictly below the value of the leftmost saturated
        # entry in the chosen starting occurrence from prop 5.
        for new_values_tuple in itertools.combinations(range(occ_s_val), v):
            new_values = set(new_values_tuple)

```

```

old_values = [x for x in range(total_distinct) if x not in new_values]
occ = tuple(old_values[val] for val in rho)

prefix_occ = occ[:s0]
suffix_occ = occ[s0:]
prefix_len = t + s0

# Place the first s0 occurrence entries somewhere in the prefix
for occ_positions in itertools.combinations(range(prefix_len), s0):
    prefix: List[int | None] = [None] * prefix_len
    ok = True
    for j, pos in enumerate(occ_positions):
        val = prefix_occ[j]
        if val > pos:
            ok = False
            break
        prefix[pos] = val
    if not ok:
        continue

filler_slots = [i for i, x in enumerate(prefix) if x is None]

# Each new value must appear at least twice among the fillers
extra_copies = m - v # beyond the mandatory 2 copies of each new value
allowed_filler_values = list(range(occ_s_val))
if 2 * len(new_values) > len(filler_slots):
    continue

for extras in weak_compositions(extra_copies, len(allowed_filler_values)):
    counts = {
        val: (2 if val in new_values else 0) + extras[val]
        for val in allowed_filler_values
    }

    # check before generating actual assignments.
    bad = False
    for val, cnt in counts.items():
        if cnt:
            cap = sum(1 for pos in filler_slots if pos >= val)
            if cnt > cap:
                bad = True
                break
    if bad:
        continue

    for filler in filler_assignments(filler_slots, counts):

```

```

        pref = list(prefix)
        for pos, val in zip(filler_slots, filler):
            pref[pos] = val
        sigma = tuple(int(x) for x in pref) + suffix_occ
        if is_rho_minimal(sigma, rho):
            out.add(sigma)

    return sorted(out)

def profile_of_rho(rho: Pattern) -> Tuple[int, ...]:
    mins = generate_rho_minimals(rho)
    by_len = Counter(map(len, mins))
    lo = min(by_len)
    hi = max(by_len)
    return tuple(by_len.get(n, 0) for n in range(lo, hi + 1))

def all_patterns_upto(max_len: int) -> Iterator[Pattern]:
    for n in range(1, max_len + 1):
        yield from gen_patterns_of_length(n)

def count_patterns_upto(max_len: int) -> int:
    # c
    return sum(1 for _ in all_patterns_upto(max_len))

def _profile_worker(rho: Pattern) -> Tuple[int, ...]:
    return profile_of_rho(rho)

def distinct_profiles_upto(max_len: int, processes: int = 1, show_progress: bool = True) ->
    patterns = list(all_patterns_upto(max_len))
    total = len(patterns)
    profiles: set[Tuple[int, ...]] = set()
    started = time.time()

    if processes <= 1:
        for i, rho in enumerate(patterns, 1):
            profiles.add(profile_of_rho(rho))
            if show_progress and (i % 100 == 0 or i == total):
                elapsed = time.time() - started
                print(f"[{i}/{total}] distinct profiles so far: {len(profiles)} elapsed: {elapsed}")
    return profiles

```

```

with mp.Pool(processes=processes) as pool:
    for i, prof in enumerate(pool.imap_unordered(_profile_worker, patterns, chunksize=10)):
        profiles.add(prof)
        if show_progress and (i % 100 == 0 or i == total):
            elapsed = time.time() - started
            print(f"[{i}/{total}] distinct profiles so far: {len(profiles)} elapsed: {elapsed}")

return profiles

def main() -> None:
    parser = argparse.ArgumentParser(description="Count distinct profile tuples P(rho) for p")
    parser.add_argument("max_len", type=int, help="Compute for all patterns rho with length")
    parser.add_argument("--processes", type=int, default=1, help="Number of worker processes")
    parser.add_argument("--show-profiles", action="store_true", help="Print the distinct profiles")
    parser.add_argument("--quiet", action="store_true", help="Suppress progress output.")
    args = parser.parse_args()

    if args.max_len < 1:
        raise SystemExit("max_len must be at least 1")
    if args.processes < 1:
        raise SystemExit("--processes must be at least 1")

    t0 = time.time()
    profiles = distinct_profiles_upto(args.max_len, processes=args.processes, show_progress=True)
    elapsed = time.time() - t0

    print(f"max_len = {args.max_len}")
    print(f"distinct profile count = {len(profiles)}")
    print(f"elapsed = {elapsed:.3f} seconds")
    if args.show_profiles:
        for prof in sorted(profiles):
            print(prof)

if __name__ == "__main__":
    mp.freeze_support()
    main()

```

Also, the full list of profiles is:

```

(1,)
(1, 1)
(2,)
(2, 3)
(2, 5, 4)

```

(4,)
(4, 1)
(4, 3)
(4, 4)
(4, 6)
(4, 7)
(4, 8, 5)
(4, 8, 6)
(4, 11, 9)
(6, 5)
(6, 21, 21)
(6, 27, 49, 34)
(7,)
(7, 1)
(7, 3)
(7, 4)
(7, 6)
(7, 10, 3)
(7, 13, 9)
(7, 14, 10)
(7, 15, 11)
(7, 16, 8)
(7, 16, 12)
(7, 17, 13)
(11,)
(11, 15, 4)
(11, 16, 5)
(11, 16, 7)
(11, 19, 12)
(12, 6)
(12, 12)
(12, 36, 31)
(12, 36, 32)
(12, 50, 86, 57)
(12, 50, 87, 59)
(12, 50, 88, 60)
(12, 56, 103, 72)
(18, 4)
(18, 26, 4)
(18, 35, 22)
(18, 45, 34)
(18, 61, 95, 58)
(18, 61, 95, 60)
(18, 92, 177, 126)
(20, 9)
(20, 72, 96, 38)

(20, 75, 121, 76)
(20, 75, 121, 78)
(20, 80, 136, 90)
(24, 54, 38)
(24, 144, 325, 264)
(24, 168, 515, 790, 496)
(34,)
(34, 104, 138, 72)
(34, 104, 152, 90)
(46, 8)
(46, 108, 69, 15)
(46, 134, 196, 120)
(48, 96, 64)
(48, 96, 66)
(48, 324, 970, 1461, 903)
(48, 324, 972, 1471, 915)
(48, 324, 974, 1477, 920)
(48, 324, 976, 1482, 924)
(72, 102, 60)
(72, 444, 1271, 1857, 1118)
(72, 444, 1271, 1866, 1134)
(72, 444, 1271, 1868, 1138)
(72, 444, 1271, 1870, 1140)
(96, 94, 16)
(96, 506, 1272, 1698, 956)
(96, 506, 1272, 1698, 960)
(120, 504, 839, 530)
(120, 1200, 5471, 13841, 18986, 11056)